Generated from docs/ac30_schematica.jpg (SWTPC AC−30 Cassette Tape Switching Schematic), docs/AC30−BOM.md and docs/ac30_wiring1a.jpg. Each block is wired; net labels join the blocks and sheets (identical labels are connected).

Transcription notes / assumptions:
- Drawing pin−number slips corrected to the real pinouts: IC7D drawn 8−>9 (4049: 9−>10); IC13C drawn inputs 7,8 (4070: 8,9).
- 4053 INH (pin 6) and VEE (pin 7) are not shown on the drawing; tied to GND.
- IC10 pins 4/8 tie to the R38 supply node, assumed +5V.
- IC15 (1488) supply pins are not shown; taken as +13V / −13V from the power supply (sheet 3).
- CONTROL INVERT: drawn as R42 pull−up; it is also brought out on J4.
- J3−J5 pin positions match the original artwork (docs/ac30_redblue150.jpg).
- The front panel is wired to five rows of 0.1 in through−hole pads instead of the artwork's J1/J2 harness headers: J1 SWITCHES, J2 LEDS, J13 MOTOR, J14 MIC (MIC A/B jacks) and J15 EAR (EAR A/B jacks). The S2/S3 audio poles wire to the SWITCHES pads (AUDIO_OUT, MIC_A/B, AUDIO_IN, EAR_A/B); the board joins them.
- MIC/EAR jack reference designators J6−J9 are assigned arbitrarily; J10/J11 are the motor jacks.
- Trimmer R39 (DELAY, next to IC10 on the artwork) is not on any schematic; it is listed, not placed, in tools/netlist.py.
- The artwork's J2 pins 11/12 (18 VAC A/B) are gone: the power supply is external.

SMT variant (kicad−smt/AC30_SMT): every on−board part is surface mount except the Molex KK−396 connectors (J3−J5, J12), the front−panel jumper pads and the mounting holes. 0805 R/C, SOD−123 diodes (1N4148W, BZT52C4V7), SOT−23 transistors (MMBT5088 for 2N5210, MMBT5087), SOIC ICs, tantalum EIA−3216 (C4, C17), 6.3x7.7 mm aluminium C16, Bourns 3314G trimmer R16, 1206 LED D18. RLY1/RLY2 are Omron G6K−2F−Y (5 V coil, DPDT 1 A, both poles in parallel as the motor contact) instead of 6 V SIL reed relays; check the coil polarity mark (pin 1 to RELAY_x).

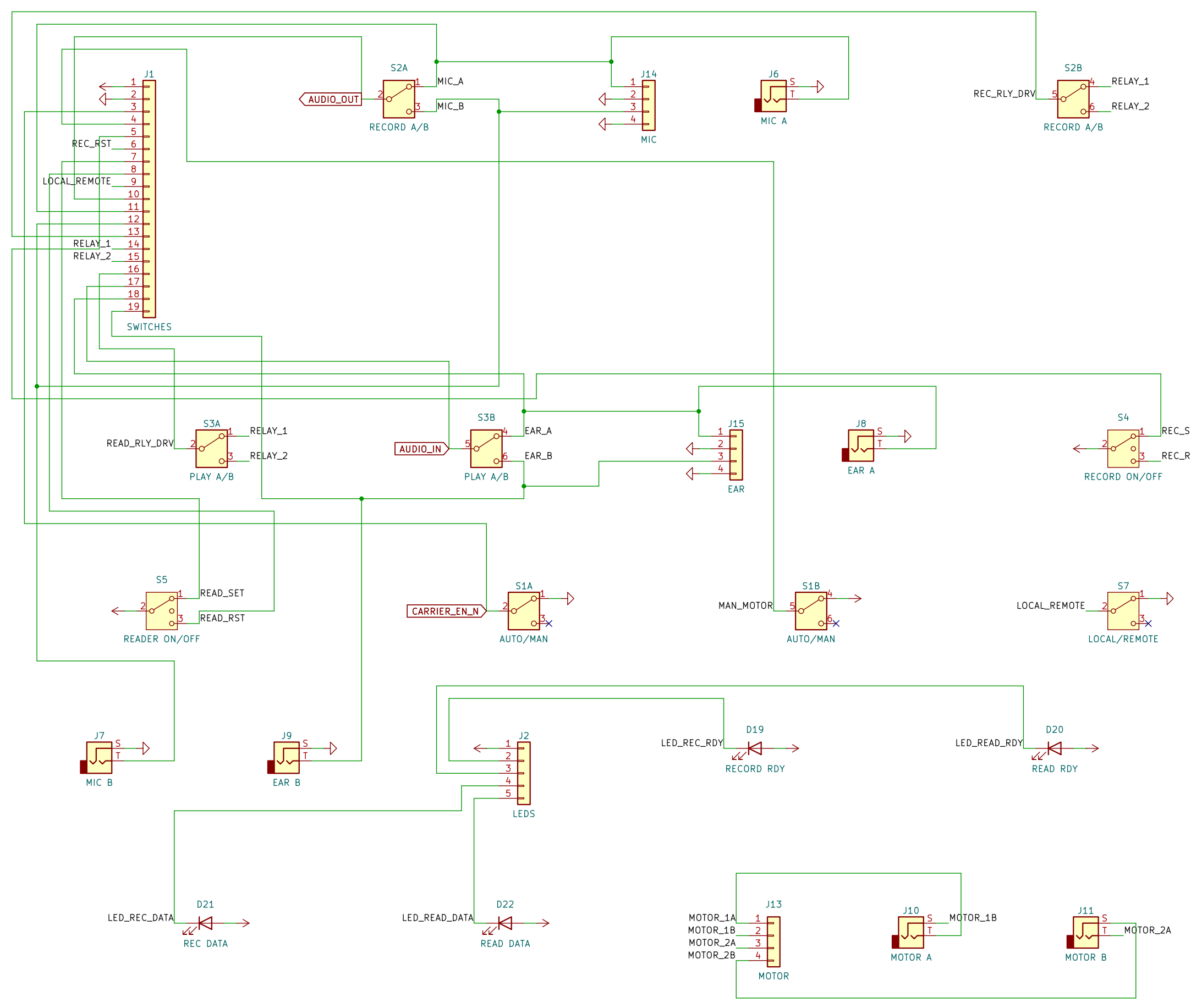
Mod/Demod & power input

File: AC30_SMT_moddemod_power.kicad_sch

Power supply (external)

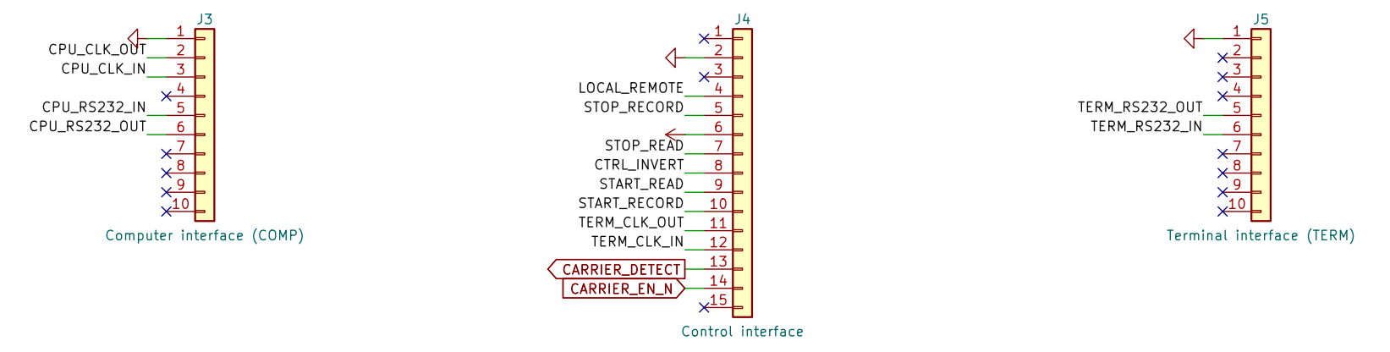
File: AC30_SMT_power_supply.kicad_sch

Front panel switches, LEDs and jacks



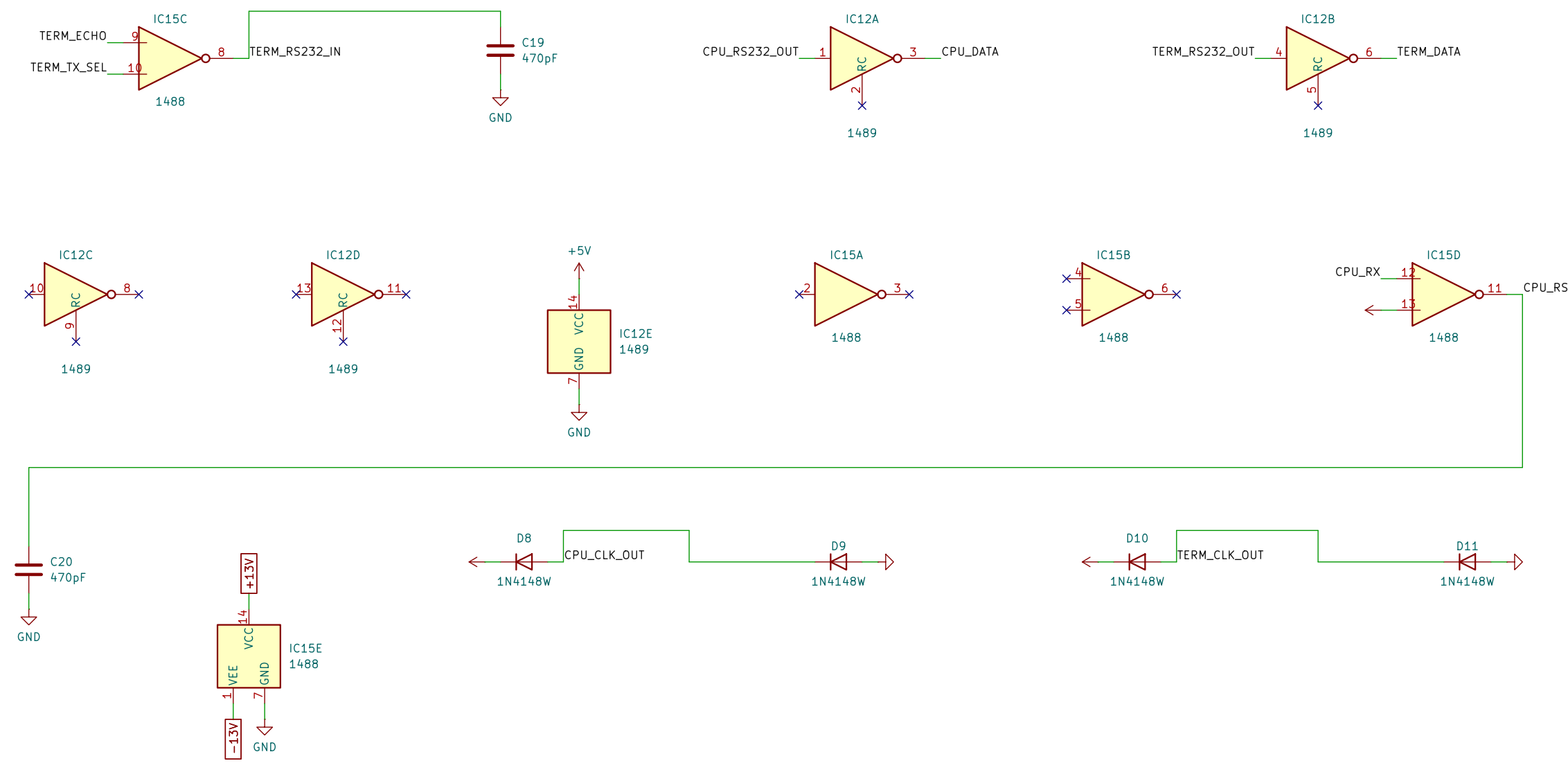
J1
SWITCHES
S2A
RECORD A/B
AUDIO_OUT
MIC_A
MIC_B
J14
MIC
J6
MIC A
S2B
RECORD A/B
REC_RLY_DRV
RELAY_1
RELAY_2
S3A
PLAY A/B
S3B
PLAY A/B
AUDIO_IN
EAR_A
EAR_B
J15
EAR
J8
EAR A
S4
RECORD ON/OFF
REC_SET
REC_RST
S5
READER ON/OFF
READ_SET
READ_RST
S1A
AUTO/MAN
CARRIER_EN_N
S1B
AUTO/MAN
MAN_MOTOR
S7
LOCAL/REMOTE
J7
MIC B
J9
EAR B
J2
LEDS
D19
RECORD RDY
LED_REC_RDY
D20
READ RDY
LED_READ_RDY
D21
REC DATA
LED_REC_DATA
D22
READ DATA
LED_READ_DATA
J13
MOTOR
MOTOR_1A
MOTOR_1B
MOTOR_2A
MOTOR_2B
J10
MOTOR A
J11
MOTOR B

Board connectors (pin positions per original artwork)



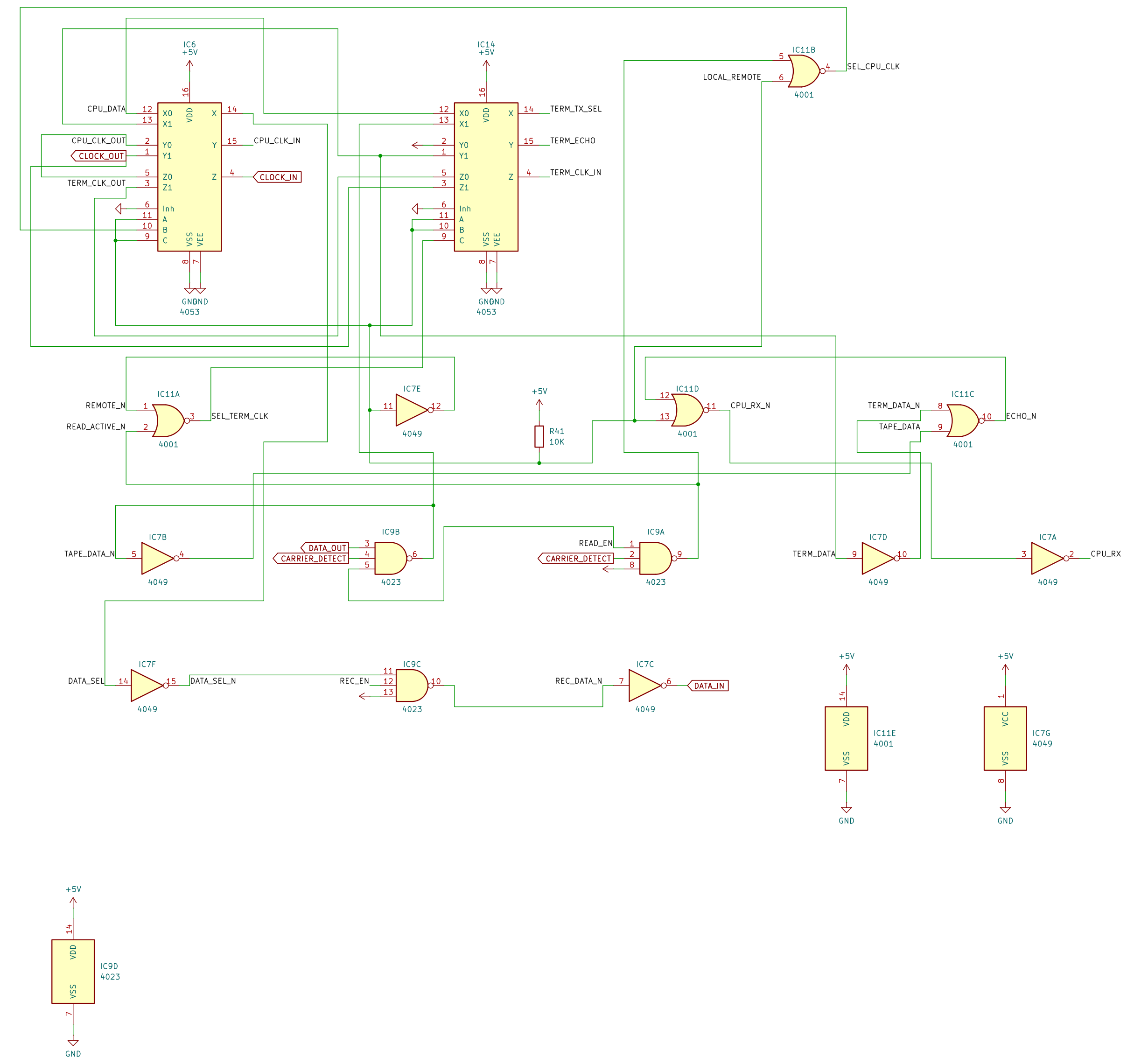
CPU_CLK_OUT
CPU_CLK_IN
CPU_RS232_IN
CPU_RS232_OUT
Computer Interface (COMP)
LOCAL_REMOTE
STOP_RECORD
STOP_READ
CTRL_INVERT
START_READ
START_RECORD
TERM_CLK_OUT
TERM_CLK_IN
CARRIER_DETECT
CARRIER_EN_N
Control Interface
TERM_RS232_OUT
TERM_RS232_IN
Terminal Interface (TERM)

RS−232 receivers / drivers, clock clamps



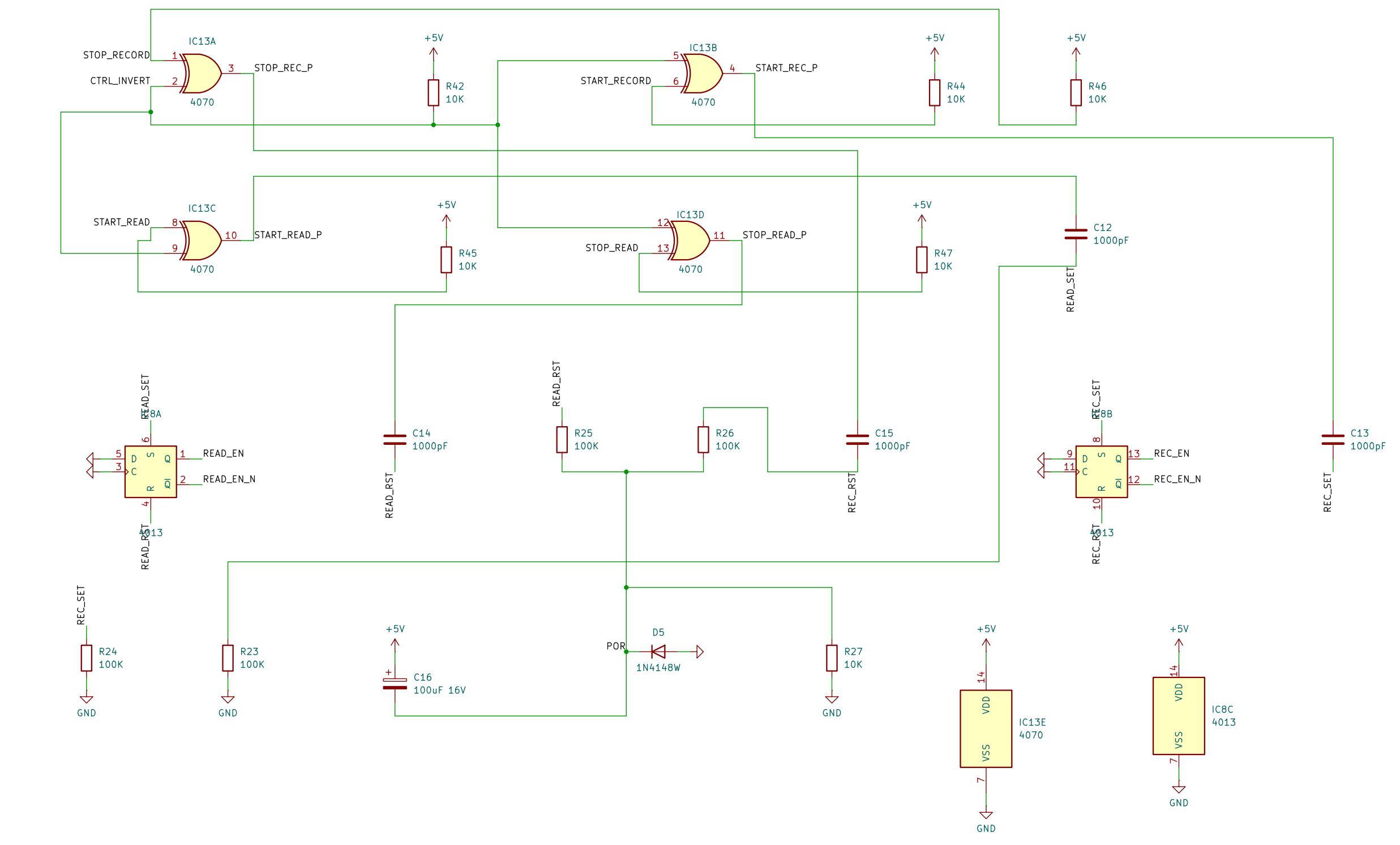
TERM_ECHO
TERM_TX_SEL
TERM_RS232_IN
1488
C19
470pF
GND
CPU_RS232_OUT
CPU_DATA
1489
TERM_RS232_OUT
TERM_DATA
IC15
1488
+5V
CPU_RX
CPU_RS232_IN
C20
470pF
D8
CPU_CLK_OUT
D4
D15
TERM_CLK_OUT
D13
1N4148W

Local/Remote steering and data/clock selectors



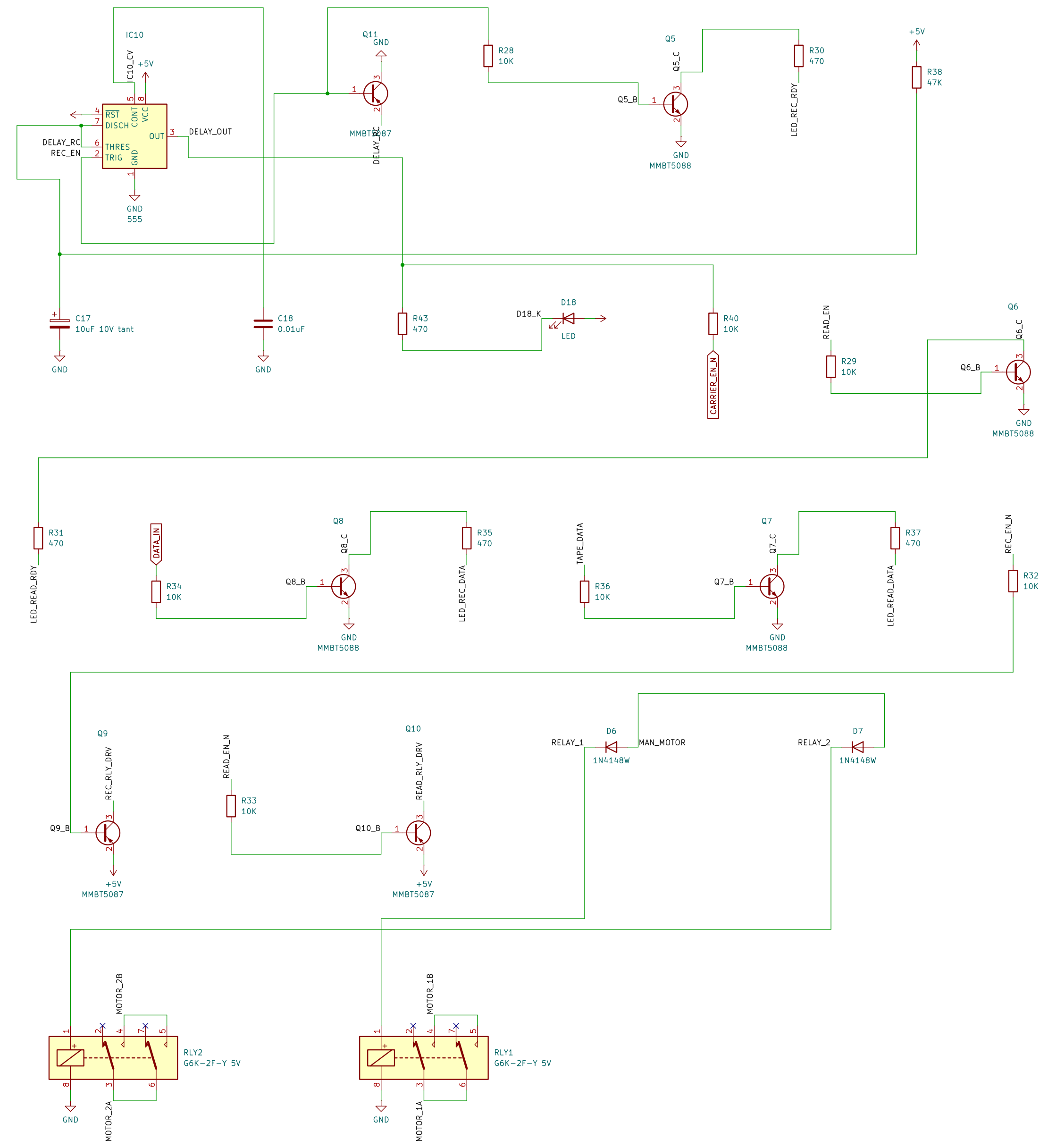
IC6
4053
IC14
4053
CPU_DATA
CPU_CLK_OUT
CLOCK_OUT
TERM_CLK_OUT
CPU_CLK_IN
CLOCK_IN
TERM_TX_SEL
TERM_ECHO
TERM_CLK_IN
LOCAL_REMOTE
SEL_CPU_CLK
4001
REMOTE_N
READ_ACTIVE_N
SEL_TERM_CLK
4049
+5V
CPU_RX_N
TERM_DATA_N
TAPE_DATA
ECHO_N
DATA_OUT
CARRIER_DETECT
4023
READ_EN
CPU_RX
DATA_SEL
DATA_SEL_N
REC_EN
REC_DATA_N
DATA_IN
GND

Record / read control latches and power−on reset



STOP_RECORD
CTRL_INVERT
STOP_REC_P
START_RECORD
START_REC_P
START_READ
START_READ_P
STOP_READ
STOP_READ_P
4070
+5V
READ_SET
READ_RST
REC_SET
REC_RST
READ_EN
READ_EN_N
REC_EN
REC_EN_N
1000pF
100K
10K
GND
4013
4070

LED drivers, relay drivers, carrier−enable delay

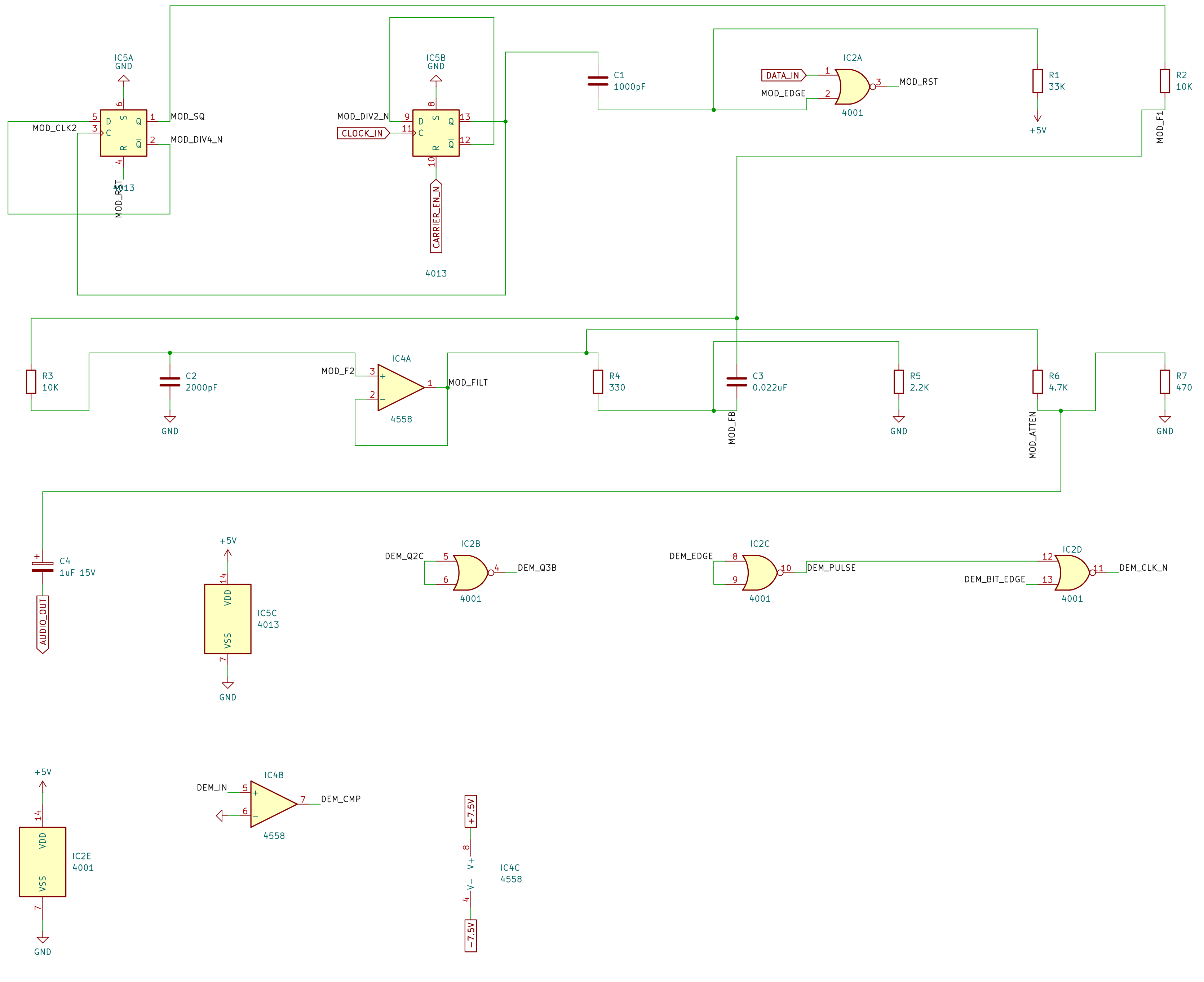


IC10
555
DELAY_RC
REC_EN
RELAY_OUT
C17
10uF 10V tant
C18
0.01uF
GND
MMBT5087
MMBT5088
LED_REC_RDY
CARRIER_EN_N
READ_EN
LED_READ_RDY
LED_REC_DATA
TAPE_DATA
LED_READ_DATA
REC_DATA_N
REC_RLY_DRV
READ_EN_N
RELAY_1
RELAY_2
MAN_MOTOR
1N4148W
RLY2
G6K−2F−Y 5V
RLY1
MOTOR_2B
MOTOR_2A
MOTOR_1B
MOTOR_1A
+5V

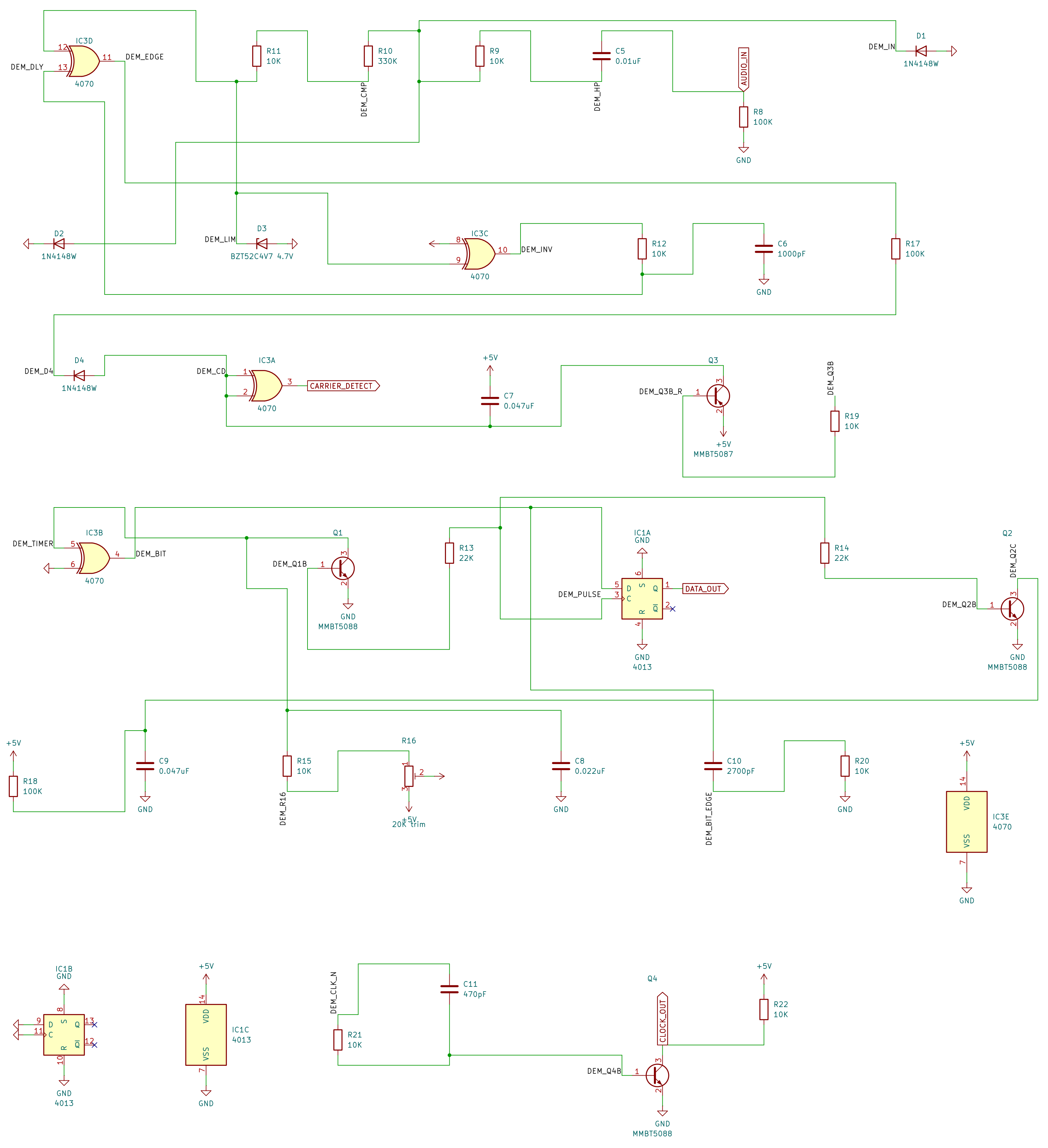
Generated from docs/ac30_mod_demod.pdf (SWTPC AC−30 Cassette Tape Modulator/Demodulator
Schematic). Each block is wired; net labels join the blocks and sheets (identical labels are connected).
Transcription notes / assumptions:
- IC4 (4558) supply pins are not drawn. The assembly manual says IC4 runs from the +/−7.5 V zener rails (D12/D13 on the power supply sheet).
- R16's far end has a terminal dot with no label; taken as +5V (Q1/C8 timer pull−up).
- IC1B is unused and not drawn: its inputs are tied to GND, its outputs left open.
- The manual's text calls the pulse inverter IC2A; the schematic (followed here) uses IC2C.

Power input: the supply (sheet 3) is built as a separate unit. Its six DC outputs come onto
the board on J12, a 6−pin KK−396 header (1 GND, 2 +5V, 3 +13V, 4 −13V, 5 +7.5V, 6 −7.5V). C23−C25 are +5 V bypass capacitors on the board
(drawn at the regulator on docs/ac30_PS.pdf; next to IC1, IC7 and IC13 on the original artwork).

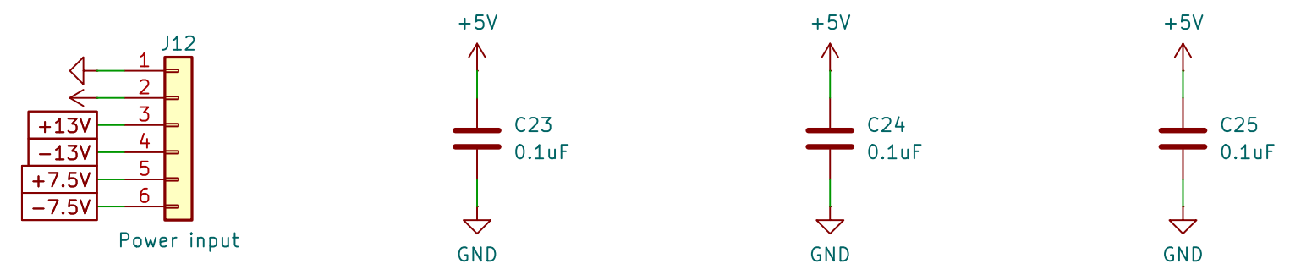
Modulator: CLOCK IN /2 (IC5B), /1 or /2 by DATA IN (IC5A), 2−pole filter IC4A


IC5A
GND
MOD_CLK2
MOD_SQ
MOD_DIV4_N
MOD_RST
4013
IC5B
GND
MOD_DIV2_N
CLOCK_IN
CARRIER_EN_N
4013
C1
1000pF
IC2A
DATA_IN
MOD_EDGE
MOD_RST
4001
R1
33K
+5V
R2
10K
MOD_F1
R3
10K
C2
2000pF
GND
IC4A
MOD_F2
MOD_FILT
4558
R4
330
C3
0.022uF
MOD_FB
R5
2.2K
GND
R6
4.7K
MOD_ATTEN
R7
470
GND
C4
1uF 15V
AUDIO_OUT
+5V
IC5C
4013
VDD
VSS
GND
IC2B
DEM_Q2C
DEM_Q3B
4001
IC2C
DEM_EDGE
DEM_PULSE
4001
IC2D
DEM_BIT_EDGE
DEM_CLK_N
4001
+5V
IC2E
4001
VDD
VSS
GND
IC4B
DEM_IN
DEM_CMP
4558
+7.5V
V+
V−
−7.5V
IC4C
4558

Demodulator: comparator IC4B, edge pulses IC3C/D, carrier detect, data and 16X clock out


IC3D
DEM_DLY
DEM_EDGE
4070
R11
10K
R10
330K
DEM_CMP
R9
10K
C5
0.01uF
DEM_HP
AUDIO_IN
R8
100K
GND
DEM_IN
D1
1N4148W
D2
1N4148W
DEM_LIM
D3
BZT52C4V7 4.7V
IC3C
DEM_INV
4070
R12
10K
C6
1000pF
GND
R17
100K
DEM_D4
D4
1N4148W
DEM_CD
IC3A
CARRIER_DETECT
4070
+5V
C7
0.047uF
Q3
DEM_Q3B_R
+5V
MMBT5087
DEM_Q3B
R19
10K
DEM_TIMER
IC3B
DEM_BIT
4070
Q1
DEM_Q1B
GND
MMBT5088
R13
22K
IC1A
GND
DEM_PULSE
DATA_OUT
GND
4013
R14
22K
Q2
DEM_Q2C
DEM_Q2B
GND
MMBT5088
+5V
R18
100K
C9
0.047uF
GND
R15
10K
DEM_R16
R16
+5V
20K trim
C8
0.022uF
GND
C10
2700pF
DEM_BIT_EDGE
R20
10K
GND
+5V
IC3E
4070
VDD
VSS
GND
IC1B
GND
GND
4013
+5V
IC1C
4013
VDD
VSS
GND
DEM_CLK_N
R21
10K
C11
470pF
Q4
CLOCK_OUT
DEM_Q4B
GND
MMBT5088
+5V
R22
10K

Power input connector (from the external supply) and +5 V bypass

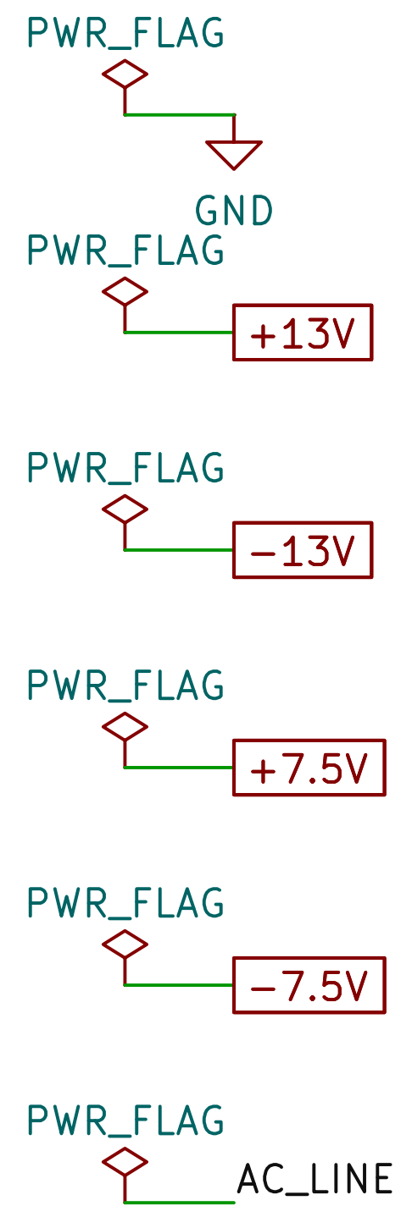

J12
+13V
−13V
+7.5V
−7.5V
Power input
+5V
C23
0.1uF
GND
+5V
C24
0.1uF
GND
+5V
C25
0.1uF
GND

Mounting holes

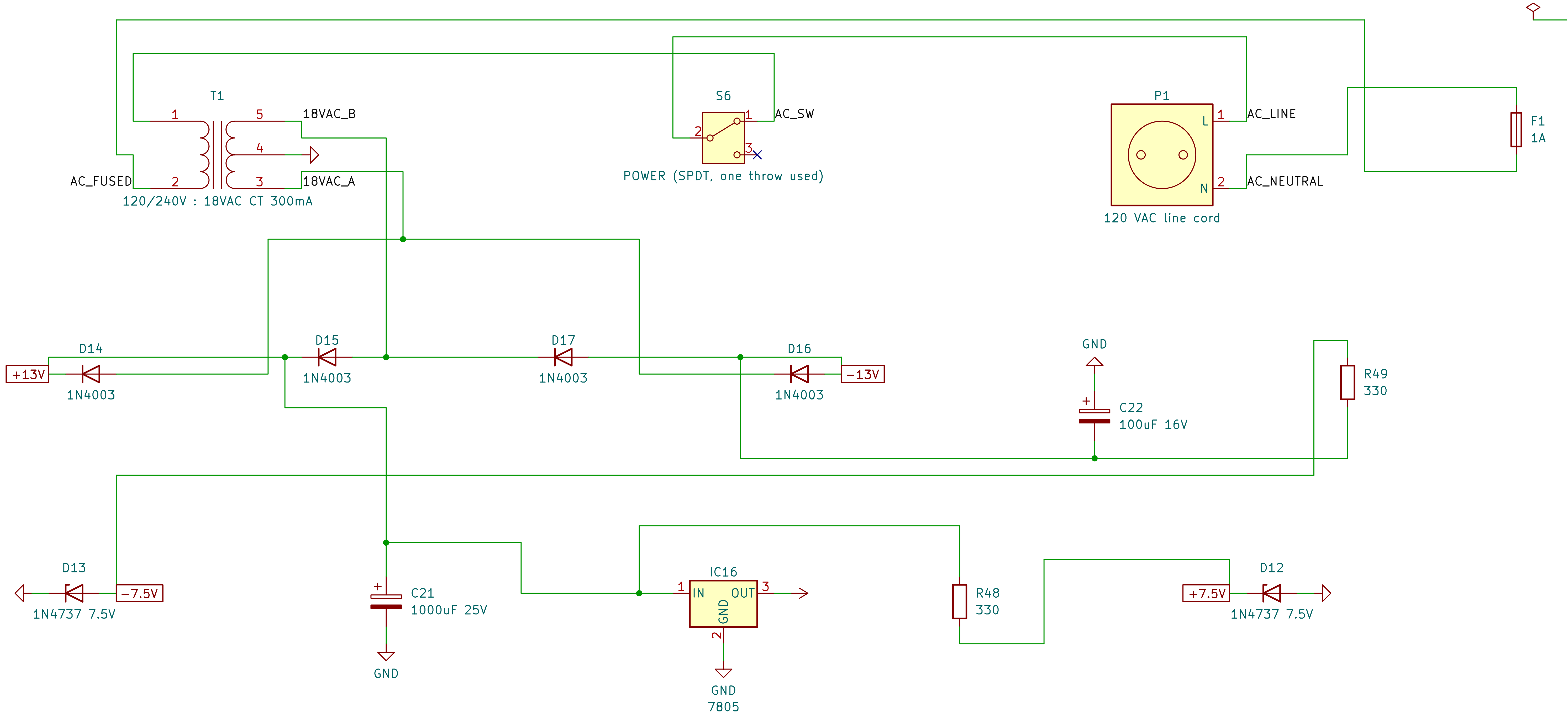
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole

Generated from docs/ac30_PS.pdf (SWTPC AC−30 Audio Cassette Power Supply Schematic).
Built as an EXTERNAL unit: these parts have no footprints and are not on the PC board. Wire the
six outputs to the board's power input J12 (KK−396: 1 GND, 2 +5V, 3 +13V, 4 −13V, 5 +7.5V, 6 −7.5V).
Transcription notes / corrections:

- The drawing joins the IC16 GND / C23−C25 return line to the bridge's negative corner (−13 V) through two crossings. That would put the 7805's ground at −13 V, so IC16 GND is taken as GND.
- T1's two 120 V primaries are drawn in parallel (120 VAC); drawn here as one primary. The 18 VAC secondary is centre−tapped, tap to GND: the bridge gives +13 V (C21) and −13 V (C22).
- D14−D17 numbering within the bridge is not shown on the drawing; assigned here.
- S6 is an SPDT toggle (BOM) used as an on/off switch; one throw is unused.
- IC15 (1488) supply pins are not drawn on the switching schematic; taken as +/−13 V, the only load for those outputs. IC4 (4558) uses +/−7.5 V (assembly manual).


PWR_FLAG
GND
PWR_FLAG
+13V
PWR_FLAG
−13V
PWR_FLAG
+7.5V
PWR_FLAG
−7.5V
PWR_FLAG
AC_LINE

Power supply (external): 120 VAC −> T1 −> bridge D14−D17 −> +/−13 V, zeners +/−7.5 V, 7805 +5 V


T1
1
5
18VAC_B
4
AC_FUSED
2
3
18VAC_A
120/240V : 18VAC CT 300mA
S6
2
1
3
AC_SW
POWER (SPDT, one throw used)
P1
L
1
AC_LINE
N
2
AC_NEUTRAL
120 VAC line cord
F1
1A
D14
+13V
1N4003
D15
1N4003
D17
1N4003
D16
−13V
1N4003
GND
C22
100uF 16V
R49
330
D13
−7.5V
1N4737 7.5V
C21
1000uF 25V
GND
IC16
1
IN
OUT
3
GND
2
GND
7805
R48
330
D12
+7.5V
1N4737 7.5V